

So, 79 cannot be written as the sum of 18 fourth powers. $\square$

Example 9: What is wrong with this "proof"?

"Theorem": If x is a real number, then x^2 is a positive real number.

"Proof": Let p_1 be " x is positive," let p_2 be " x is negative," and let q be " x^2 is positive." To show that $p_1 \rightarrow q$ is true, note that when x is positive, x^2 is positive because it is the product of two positive numbers, x and x . To show that $p_2 \rightarrow q$, note that when x is negative, x^2 is positive because it is the product of two negative numbers, x and x . This completes the proof.

Solution: The problem with this "proof" is that we missed the case of $x=0$. When $x=0$, $x^2=0$ is not positive, so the supposed theorem is false. If p is " x is a real number," then we can prove results where p is the hypothesis with three cases, p_1, p_2 and p_3 , where p_1 is " x is positive," p_2 is " x is negative," and p_3 is " $x=0$ " because of the equivalence $p \leftrightarrow p_1 \vee p_2 \vee p_3$. $\square$

1.8.3 Existence Proofs

Many theorems are assertions that objects of a particular type exist. A theorem of this type is a proposition of the form $\exists x P(x)$, where P is a predicate. A proof of a proposition of the form $\exists x P(x)$ is called an existence proof.

Sometimes an existence proof of $\exists x P(x)$ can be given by finding an element a , called a witness, such that $P(a)$ is True. This type of existence proof is called constructive.

It is also possible to give an existence proof that is nonconstructive; that is, we do not find an element a such that $P(a)$ is true, but rather prove that $\exists x P(x)$ is true in some other way.

One common method of giving a nonconstructive existence proof is to use proof by contradiction and show that the negation of the existential quantification implies a contradiction. $\neg \exists x P(x) \rightarrow \text{contradiction}$